PINGSHI (JACOB) YU

Research Scientist / Research Engineer — Programming Languages and Software Reliability

London, UK p.yu22@imperial.ac.uk pingshiyu.github.io

GitHub: pingshiyu LinkedIn Google Scholar

PROFILE

Final-year Computer Science PhD researcher combining programming-languages theory with practical systems implementation. In my research, I build methods and tools for finding compiler bugs and reasoning about software correctness. This work spans fuzzing, composable compiler semantics, property-based testing, complexity theory, and automated verification. Prior industry experience includes programming-language research, quantitative engineering for an FX market-making platform, quantum software, and data engineering.

EDUCATION

Imperial College London | *PhD in Computer Science* 2022–present

Full doctoral scholarship. Research on compiler testing, composable semantics, property-based testing, and the complexity of randomised testing. Supervised by Prof. Alastair F. Donaldson and Prof. Nicolas Wu.

University of Oxford, St Anne's College | *MMathCompSci* 2016–2020

Mathematics and Computer Science; first-class master's year (80.3%). Dissertation: *Systematic Pruning Methods for the Enumeration of Realistic Molecules*.

RESEARCH HIGHLIGHTS

Compiler Testing and Composable Semantics — Imperial College London

- Developed *Ratte*, a semantics-driven approach to fuzzing multi-level compilers for miscompilations; published and presented at ASPLOS 2025.
- Researched effect-handler-based semantics for composable compiler IRs, connecting executable language semantics with automated compiler validation.
- Contributed to *RustSmith*, a random differential testing tool for the Rust compiler, published at ISSTA 2023.

Theory of Randomised Testing — Imperial College London

- Developed a complexity-theoretic account of randomised testing, linking practical testing strategies with formal questions about bug-finding difficulty; released on arXiv in 2026.

Effect Handlers for Cangjie — Huawei Research

- Researched novel applications of effect handlers—a programming-language feature—for software testing in Huawei's open-source Cangjie programming language.

INDUSTRY AND RESEARCH EXPERIENCE

Huawei Research — Edinburgh, UK

Research Intern, Programming Languages Lab 2025–2026

- Researched novel applications of effect handlers (a programming-language feature) for software testing in Huawei's open-source Cangjie programming language.

Credit Suisse — London, UK

Software Engineer / Quantitative Strategist 2020–2022

- Developed components of a production foreign-exchange market-making platform.
- Led delivery of NLP-derived market-data pipelines into the pricing engine, coordinating technical direction across quantitative, support, and data teams.
- Built services for quantitative research, model validation, release management, performance monitoring, and developer productivity.
- Worked with Python, C++, Docker, Jenkins, JavaScript, Boost, SQL, and Elasticsearch.

Cambridge Quantum Computing (now Quantinuum) — Cambridge, UK

Research Intern 2020

- Surveyed ZX-calculus research and implemented a scalable graphical calculus in C++ for representing, reasoning about, and simplifying large quantum circuits.

Earlier Engineering and Data Experience 2017–2019

- **Credit Suisse:** delivered two production projects and identified an optimisation yielding approximately $900\times$ faster bulk SQL inserts in an internal Perl API.
- **Arm:** initiated and delivered the automated data-processing backend for a company-wide capacity-planning project.
- **University of Oxford DPIR:** built a Python pipeline to process roughly 10,000 statutory instruments and designed validation materials for sentiment analysis.

PUBLICATIONS

1. **Pingshi Yu**, Chengsong Tan, Nicolas Wu, and Alastair F. Donaldson. "Complexity Theory of Randomised Testing." arXiv, 2026. Paper [link]
2. **Pingshi Yu**, Nicolas Wu, and Alastair F. Donaldson. "Ratte: Fuzzing for Miscompilations in Multi-Level Compilers Using Composable Semantics." *ASPLOS*, 2025. DOI [link]
3. Mayank Sharma, **Pingshi Yu**, and Alastair F. Donaldson. "RustSmith: Random Differential Compiler Testing for Rust." *ISSTA*, 2023. DOI [link]
4. **Pingshi Yu**. "Reasoning about MLIR Semantics through Effects and Handlers." *ECOOP/ISSTA Doctoral Symposium*, 2023. DOI [link]
5. **Pingshi Yu**, Alistair J. Sterling, and Jotun Hein. "A Novel Automated Screening Method for Combinatorially Generated Small Molecules." *Journal of Chemical Information and Modeling*, 2021. DOI [link]

TECHNICAL SKILLS

Programming	Haskell, Python, C++; experience with Agda, Java, Scala, Perl, and MATLAB.
Research areas	Compilers, fuzzing, property-based testing, programming-language semantics, effect systems and handlers, algorithms, automated verification, complexity theory.
Engineering	Git, Unix, Docker, Jenkins, SQL, Boost, Pandas, TensorFlow, Elasticsearch, LaTeX.
Languages	Chinese (native), English (native), German (basic).

RECOGNITION

- Full Doctoral Scholarship, Imperial College London, 2022–2027.
- Poster Presentation Award, Imperial Computing Conference, 2024.
- Invited research presentation at ETH Zürich, 2024; paper presentations at ASPLOS 2025 and ISSTA 2023.
- Exhibition Awards, St Anne's College, University of Oxford, 2019 and 2020.

Last updated August 2026